Practical Applications of Discourse Relations

Lecture slides: Computational Linguistics 2 | CL3.202

LANGUAGE TECHNOLOGIES
RESEARCH CENTER

INTERNATIONAL INSTITUTE OF
INFORMATION TECHNOLOGY
HYDERABAD

What was the latest media you've seen?

- Podcast (2)
- Youtube video (7)
- Youtube short/tiktok (12)
- Comment thread (4)
- news (3)
- books (12)

- Slides (no, let us not make the graph skewed, perfectly valid tho)
- Whatsapp (no, lets stick to public media) :P

- Hey GPT, convert it into a bar plot!

Class objectives

- 4 questions that will be graded. Last 15 mins for writing.
- Topics will be part of exams./?

Take home assignment

- What are coherence relations and markers?
- How does back translation improve implicit DR classification?
- What are the challenges in transfer learning DR across datasets?
- What is class imbalance? What strategies are used for it?

References

*Your class notes!

Computational concepts

- Jurafsky
- DiscoGem
- DiscoFlan
- GUM
- Transfer learning
- Class imbalance
- Uncertainty in classification

Major linguistic theories

- PDTB 2 manual
- PDTB 3 manual
- RST paper by Mann and Thompson
- SDRT

Motivation: DMs = Coherence ≃ *Intent

Pragmatics and Discourse

Linguistics → Pragmatics → **Discourse Analysis → Cohesion and Coherence** → Discourse Markers

- **Pragmatics** is a branch of linguistics that studies how **context** influences the way we interpret and use language. It goes beyond the literal.
  - Ellipsis
  - Anaphora Resolution
  - Discourse Relations
  - Deixis
  - Implicature
  - Referring expressions
  - Presupposition

You need context to understand these phenomena

- **DA** examines how larger units of text (discourse) are structured and how they function.
- **Coherence** refers to the overall **logical sense and unity of ideas** in a text.

Motivation: DMs = Coherence ≃ *Intent

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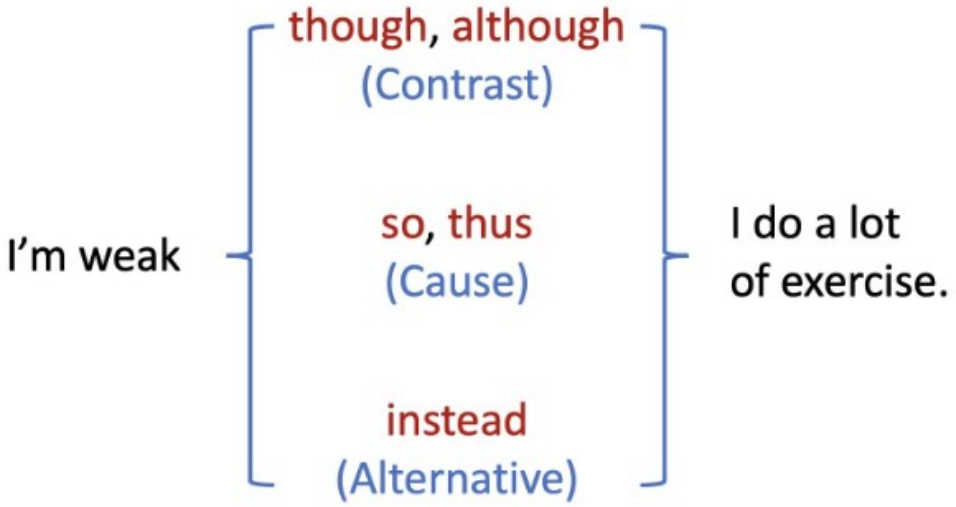

Figure 1: Entangled discourse relations and corresponding markers between clauses. As shown in the figure, there exist diverse discourse relations (marked in blue) and corresponding markers (marked in red) for the same pair of clauses. It suggests that the semantic meaning of different discourse relations can be entangled to each other.

Motivation: DMs = Coherence ≃ *Intent

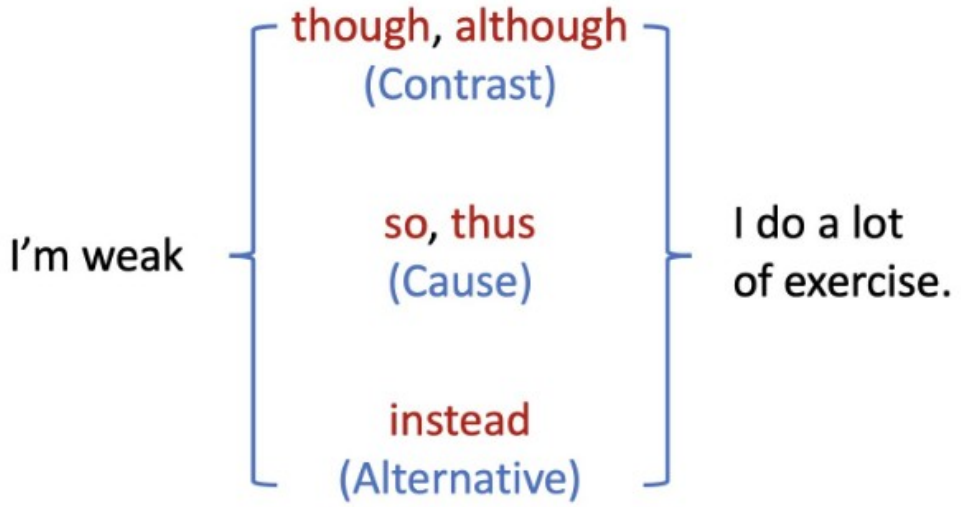

Figure 1: Entangled discourse relations and corresponding markers between clauses. As shown in the figure, there exist diverse discourse relations (marked in blue) and corresponding markers (marked in red) for the same pair of clauses. It suggests that the semantic meaning of different discourse relations can be entangled to each other.

I do a lot of exercise → something interferes with exercise (health, metabolism, poor nutrition?) → exercise doesn't improve health → I'm

I'm weak. Even **though**, I do a lot of exercise (*the contradiction*).

I'm weak → I want stay healthy → I take steps to be healthy → I do a lot of exercise

I'm weak. **So, in order to stay healthy**, I do a lot of exercise (*the result*).

I'm weak → I should be resting / taking alternative therapy → I shouldn't exercise → I am stubborn → I do a lot of

I'm weak. **Instead of resting** (*the expected action*), I do a lot of exercise (*the replacement action*).

Claude 3.7 SONNET

deepseek R1

o1

Explaining in a ***Chain-of-thought*** manner

Motivation: DMs = Coherence ≃ *Intent

मैं कमजोर हूँ, जबकि मैं बहुत व्यायाम करता हूँ।

मैं कमजोर हूँ, इसलिए मैं बहुत व्यायाम करता हूँ।

मैं कमजोर हूँ, (aur mujhe araam karna chahiye) इसके बजाय मैं बहुत व्यायाम करता हूँ।

“నేను బలహీనంగా ఉన్నాను, కానీ నేను చాలా వ్యాయామం చేస్తాను.”
(“nēnu balahīnangā unnānu, kānī nēnu chālā vyāyāmaṁ chestānu.”)

“నేను బలహీనంగా ఉన్నాను, అందుకే నేను చాలా వ్యాయామం చేస్తాను.”
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“నేను బలహీనంగా ఉన్నాను, దాని బదులుగా నేను చాలా వ్యాయామం చేస్తాను.”
(“nēnu balahīnangā unnānu, dāni badulugā nēnu chālā vyāyāmaṁ chestānu.”)

Discourse

Like movies, language does not normally consist of isolated, unrelated sentences, but instead of collocated, structured, coherent groups of sentences. We refer to such a coherent structured group of sentences as a discourse.

Discourse

8. Analysis of a larger text

Thus far, our example texts illustrating RST relations have been relatively short. In this section we will apply RST to the analysis of a larger text. In an earlier paper, we have analyzed this text in terms of relational propositions (Mann and Thompson, 1986b), and also discussed it in Thompson and Mann (1987). It is an advertisement for computer diskettes from '*BYTE*' magazine.[11]

1. What if you're having to clean floppy drive heads too often?
2. Ask for Syncom diskettes, with burnished Ectype coating and dust-absorbing jacket liners.
3. As your floppy drive writes or reads,
4. a Syncom diskette is working four ways
5. to keep loose particles and dust from causing soft errors, drop-outs.
6. Cleaning agents on the burnished surface of the Ectype coating actually remove build-up from the head,
7. while lubricating it at the same time.
8. A carbon additive drains away static electricity.
9. before it can attract dust or lint.
10. Strong binders hold the signal-carrying oxides tightly within the coating.

11a. And the non-woven jacket liner,

12. more than just wiping the surface,

11b. provides thousands of tiny pockets to keep what it collects.[12]

13. To see which Syncom diskette will replace the ones you're using now,
14. send for our free 'Flexi-Finder' selection guide and the name of the supplier nearest you.
15. Syncom, Box 130, Mitchell, SD 57301. 800-843-9862; 605-996-8200.

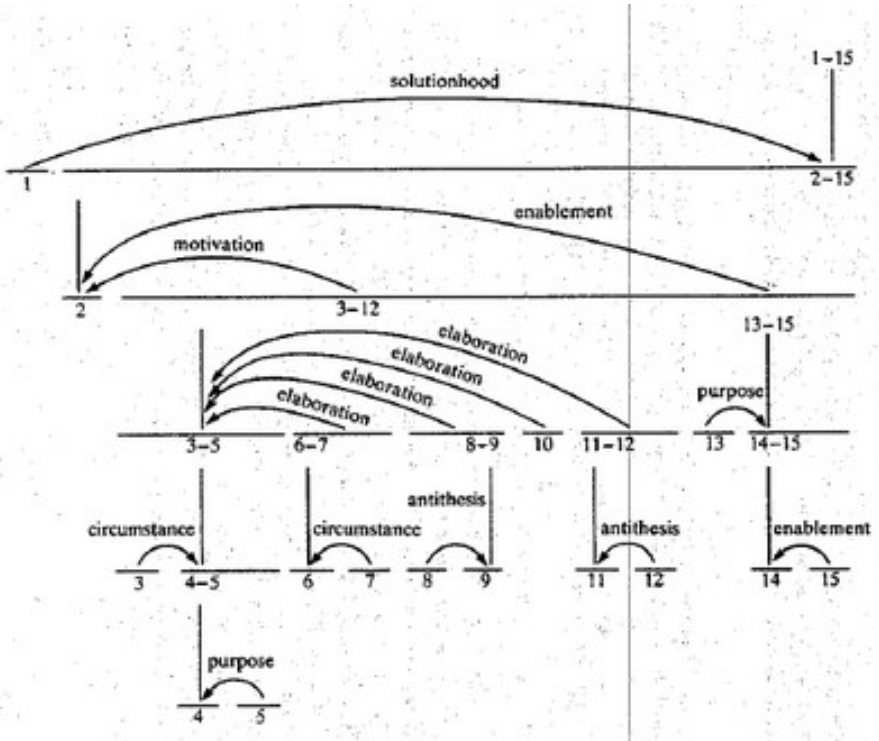

Figure 6. *RST diagram for 'Syncom' text*

Discourse

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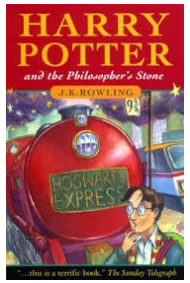
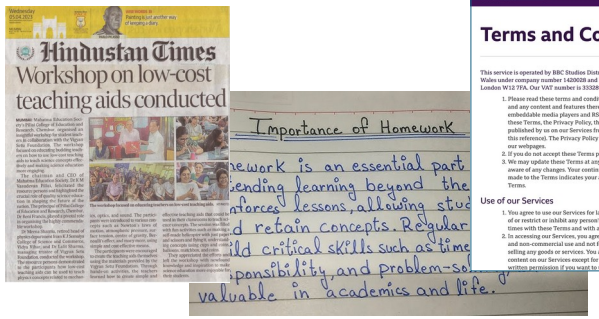
Context is Important: The meaning of discourse depends entirely on its social, cultural, and situational context. (Who is speaking/addressing whom? Where? Why?)

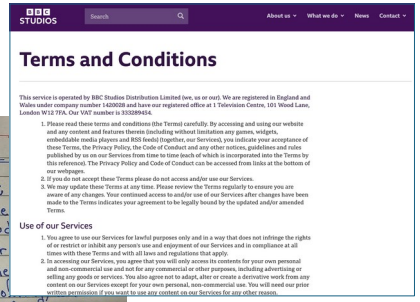
Discourse

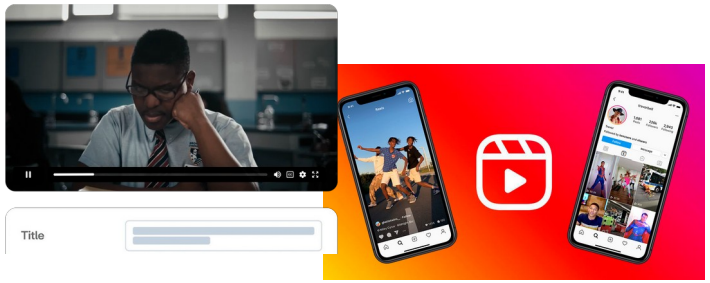
Examples of Discourse

- **Spoken:**
  - An everyday conversation with a friend.
  - Podcasts
  - A politician's speech.
  - A classroom lesson.
- **Written:**
  - A news article. (or a para from an article)
  - A thread of text messages or social media posts.
  - An academic essay.
  - The instruction manual for a new phone.
- **Visual**
  - Movies
  - A sitcom episode
  - Reels





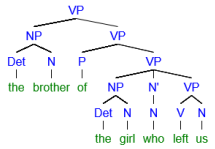


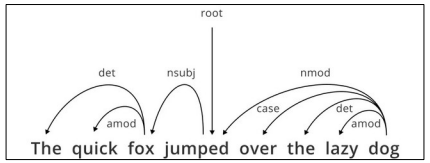


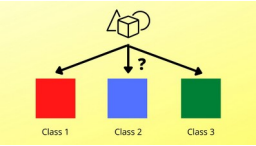
Discourse

Why study discourse?

- **Practically speaking,** we can perform more linguistic functions**. Also, the same problems** (solved with semantics and syntax) can now have a new approach **over long context.**
- **Previously** you've looked at applications like:
  1. POS tagging
  2. Syntax tree generation
  3. Small context QA
  4. Sentiment Analysis/Sentence Classification (<~512 tok)
  5. Short context dialogue bots (semantics, not prags.)







- **Now (in discourse analysis)** we can consider standard tasks like
  1. Summarization
  2. Dialogue Act tagging
  3. Implicature generation/classification
  4. Coreference resolution
- And can also have **more applications** like
  1. Long context bots
  2. Long document QA
  3. Video understanding
  4. Long doc/multi doc summarization

Discourse

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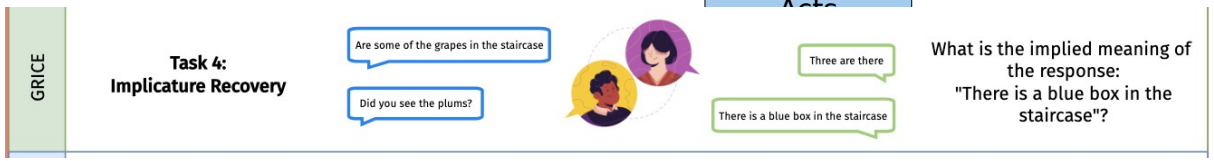
Dialogue



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Discourse

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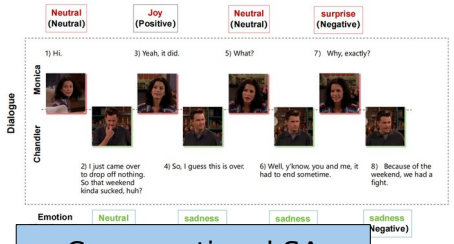
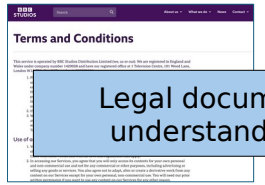
  1. Summarization
  2. Dialogue Act tagging
  3. Implicature generation/classification
  4. Coreference resolution
- And can also have **more applications** like (representnt. issues like sparsity, etc.)
  1. Long context bots
  2. Long document QA
  3. Video understanding
  4. Long doc/multi doc summarization



Conversational SA



Legal document understanding

Summarizing 1 million words

A boy discovers he is a wizard and must protect the world from the evil Lord Voldemort.

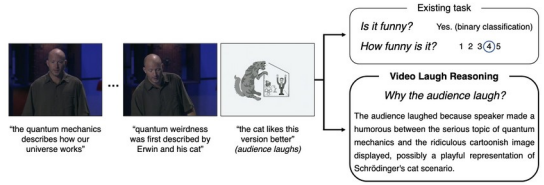
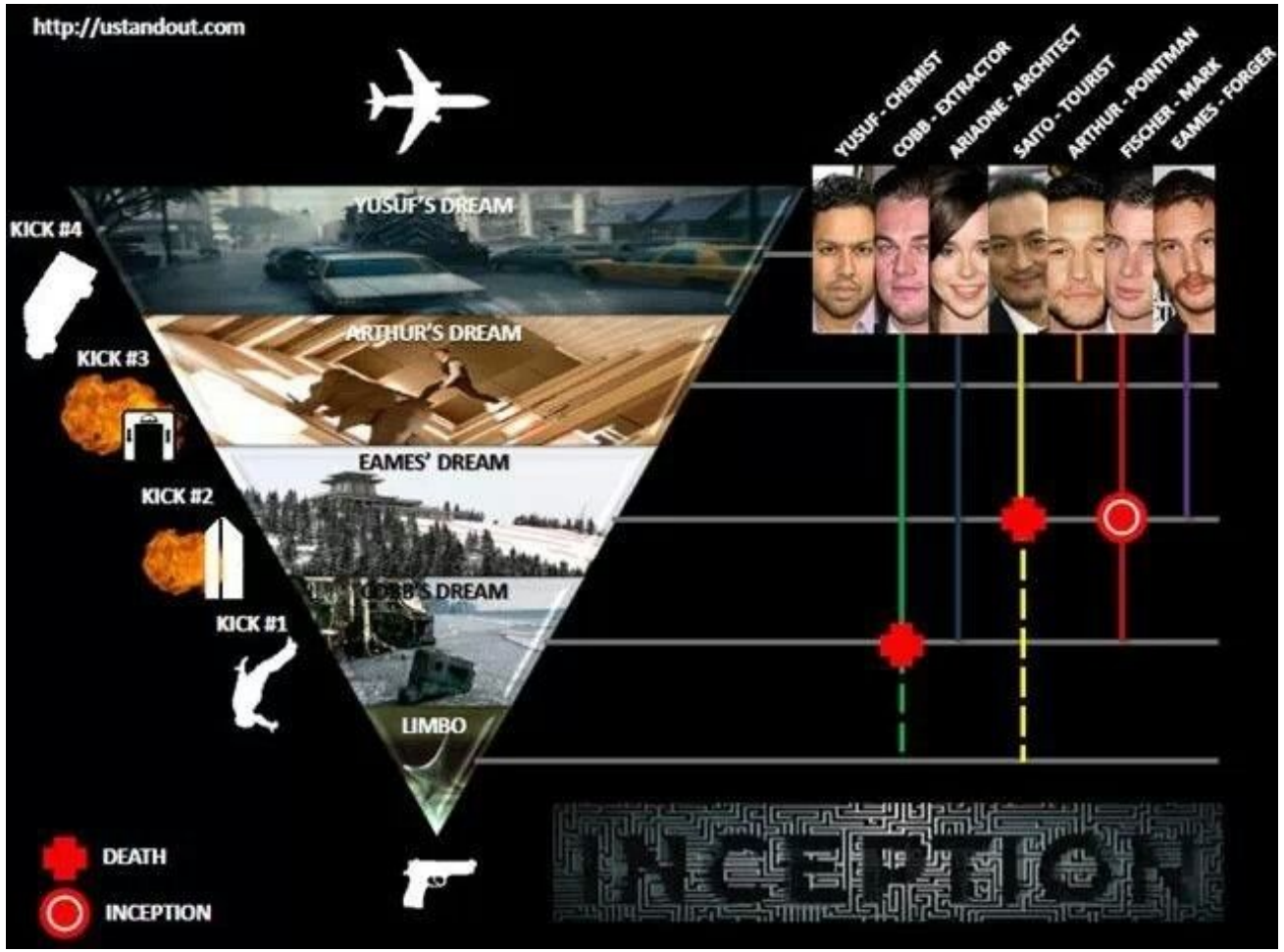
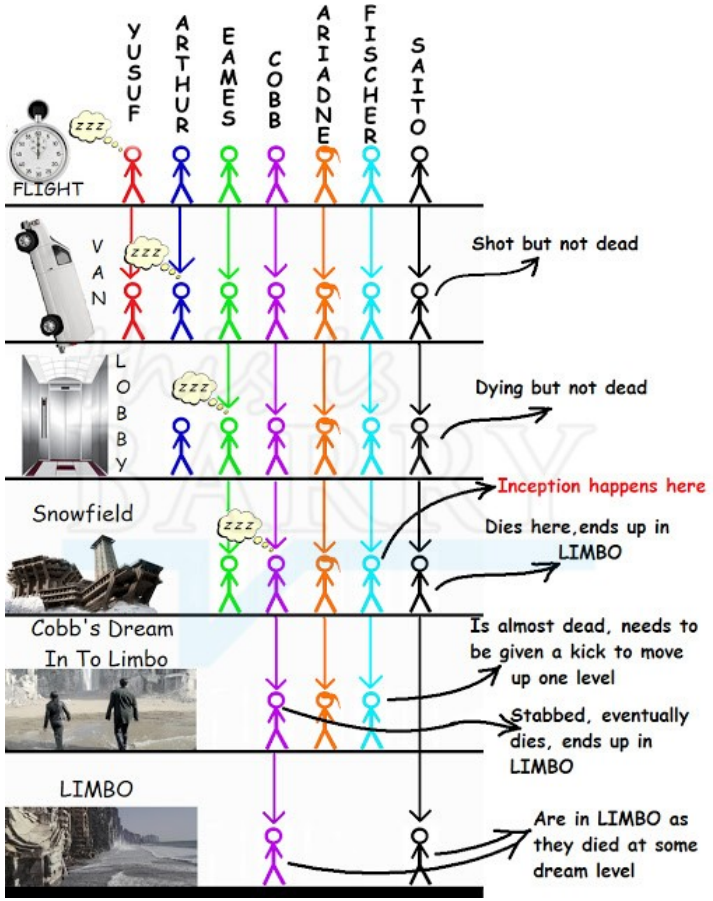


Figure 1: Why do people laugh? We present *Video Laugh Reasoning*, a new task to interpret the reasons behind laughter in a video.

Visual humour understanding

Coherence


http://ustandout.com
YUSUF - CHEMIST
COBB - EXTRACTOR
ARIADNE - ARCHITECT
SAITO - TOURIST
ARTHUR - POINTMAN
FISCHER - MARK
EAMES - FORGER
KICK #4
YUSUF'S DREAM
KICK #3
ARTHUR'S DREAM
EAMES' DREAM
KICK #2
COBB'S DREAM
KICK #1
LIMBO
DEATH
INCEPTION
INCEPTION


YUSUF
ARTHUR
EAMES
COBB
ARIADNE
FISCHER
SAITO
FLIGHT
VAN
LOBBY
Snowfield
Cobb's Dream
In To Limbo
LIMBO
Shot but not dead
Dying but not dead
Inception happens here
Dies here, ends up in LIMBO
Is almost dead, needs to be given a kick to move up one level
Stabbed, eventually dies, ends up in LIMBO
Are in LIMBO as they died at some dream level

Coherence

- Discourse has patterns. It uses **coherence** (logical flow) to make sense.

- What makes a discourse coherent?
- If you created a text by taking random sentences each from many different sources and pasted them together, would that be a coherent discourse?
- Real discourses exhibit both **local coherence** and **global coherence**.

John took a train from Paris to Istanbul. He likes spinach.

Vs.

Jane took a train from Paris to Istanbul. She had to attend a conference.

[REASON]

How do we organize longer pieces of text?

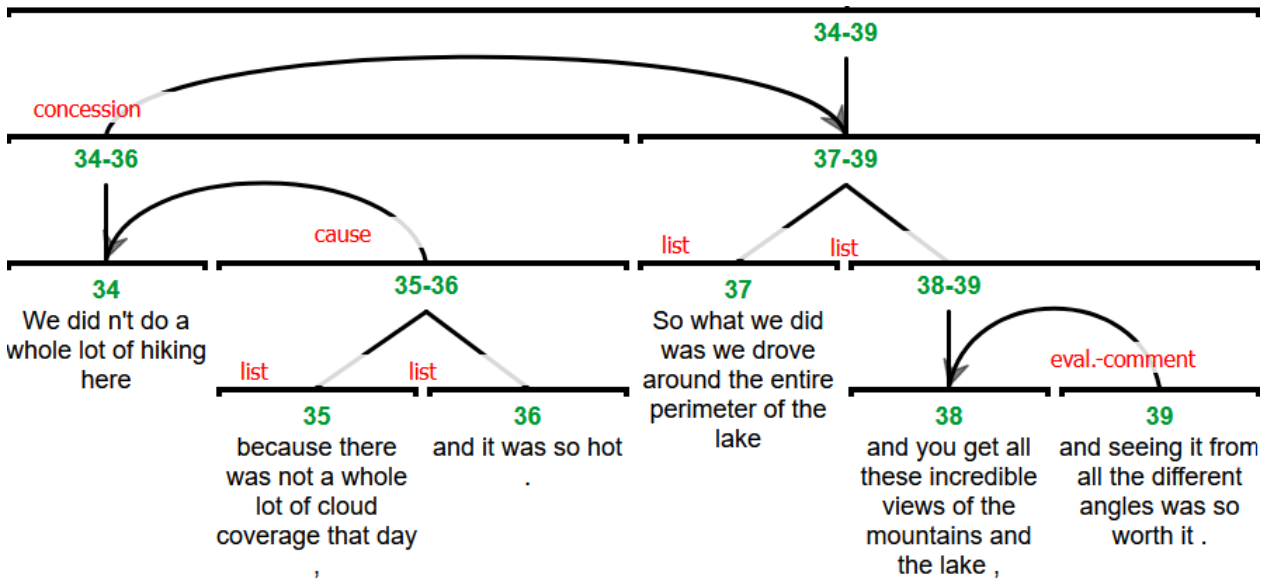
Coherence

- Why do we care about **coherence** of a discourse?

- Since coherence is a property of a well-written text, it plays a part in any task that requires measuring the quality of a text.
  - Summarization
  - Long document understanding/QA
  - Movie plot analysis
  - Essay grading
  - Stance understanding in debates/reddit threads/comments

Coherence Relations / Discourse Relations

- Structured *relationships* like REASON that hold between *text units* are called **coherence relations/ DISCOURSE RELATIONS (DR)**.
- Coherent discourses are by many such coherence/discourse relations.



How do we organize longer pieces of text?

Elementary Discourse Units

- An **Elementary Discourse Unit (EDU)** is the smallest meaningful clause or sentence that acts as a fundamental building block for analyzing discourse relations.
- *A single sentence can be made up of one or more EDUs.*

(1) The girl read the book. [EDU1]

(2) He was happy [EDU2] CAUSE
because he won the game. [EDU3]

What are the pieces in discourses?

We first break down a text into these EDUs, and *then* we analyze the relationship between them (e.g., EDU 2 is the **Cause** for EDU 1).

Frameworks for Discourse Relations

Commonly used models of coherence relations and associated corpora:

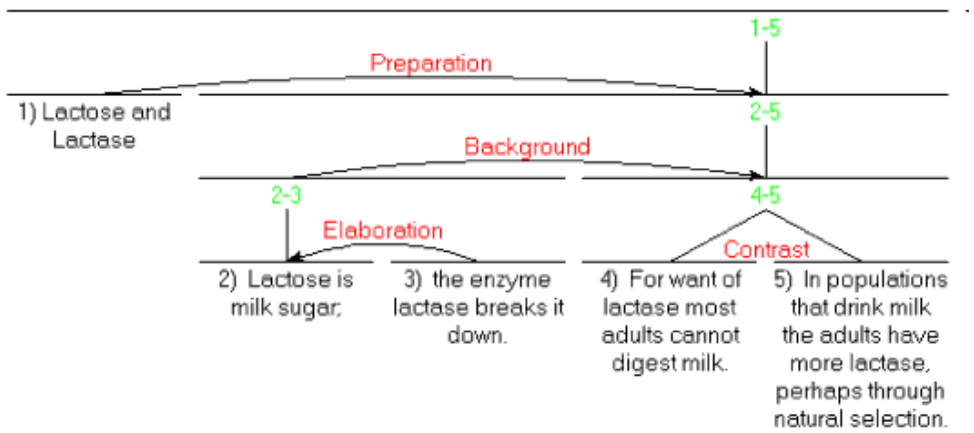
- **Rhetorical Structure Theory (RST)**
- **Penn Discourse TreeBank (PDTB)**
- Segmented Discourse Representation Theory (SDRT)

How do we organize longer pieces of text?

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

- RST analyses a discourse as a **tree** trying to understand the *structure* of the discourse.
- In RST **relations** **(Discourse Relations)** are defined between RST two spans of text (EDUs), generally a *nucleus* and a *satellite*.
- The nucleus is the unit that nucleus satellite is more **central** to the writer's purpose and that is interpretable **independently**;
- the satellite is **less central** and generally is only interpretable **with respect to the nucleus**.
- Some symmetric relations, however, hold between two nuclei.



Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

- Relations can be nucleus-satellite relations or

Relation Name	Nucleus	Satellite
Background	text whose understanding is being facilitated	text for facilitating understanding
Elaboration	basic information	additional information
Preparation	text to be presented	text which prepares the reader to expect and interpret the text to be presented.

Note: Relations are not hierarchical

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

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Relation Name	Nucleus	Satellite
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Elaboration	basic information	additional information
Preparation	text to be presented	text which prepares the reader to expect and interpret the text to be presented.

- **Multinuclear relations**: If a relation does not have a particular span of text which is more central to the author's purposes, it is called Multinuclear. An example is the neutral Contrast relation.

Relation Name	Nucleus	Satellite
Background	text whose understanding is being facilitated	text for facilitating understanding

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

- Let us jump to the RST manual to have a look at the relations.

- We need to study –
- Figure 7

- Relation definitions

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

Relation	Simplified Definition	Example
Contrast	Two statements are about a similar topic, but they highlight a key **difference** between them.	[I love sunny days] (N), but [I don't like the heat] (N).
Disjunction	Presents an **alternative or a choice** between two or more statements.	[You can have tea] (N), or [you can have coffee] (N).
Conjunction	Two or more statements of **equal importance** are simply presented together as a related unit.	[The sky is blue] (N), and [the grass is green] (N).
List	A series of comparable items or actions are presented in a *sequence*.	[You will need flour] (N), [sugar] (N), and [eggs] (N).
Volitional Cause	An event happened because a person **made a choice**, and the text explains the **reason** for it.	[He sold his car] (N) because [he needed the money] (S).
Non-Volitional Cause	An event happened **naturally or accidentally** (no choice), and the text explains the physical cause.	[The road was slippery] (N) because [it had been raining] (S).
Volitional Result	An event happened, and because of it, someone **chose to take an action**.	[The company offered him a promotion] (N). As a result, [he decided to move] (S).
Non-Volitional Result	An event happened, and it led to an outcome that **occurred automatically**.	[He fell off his bike] (N). As a result, [he broke his arm]

Rhetorical Structure Theory (RST)

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Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

Why bother with **volitionality**?

Why do we need to make a distinction between:

1. **Volitional** causes/results (involving a conscious choice).
2. **Non-volitional** causes/results (happening naturally or accidentally).

- Is this distinction just theoretical, *or does it reflect how we actually use language?*
- Many **languages require their speakers to differentiate volitionality.**
- If a concept is important for communication, it gets its own unique words.
- Dutch language provides a perfect example.

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

1. **Non-Volitional Result (Event → Event):** an event physically causes another event. No choice is involved.

English: The road was icy. **As a result**, the car slid into the ditch.

Dutch: *Het was glad op de weg.* ***Daardoor*** *gleed de auto de sloot in.*

Analysis: The ice directly *caused* the sliding. The car did not make a choice.

2. **Volitional Result (Event → Choice):** an event gives someone a *reason* to make a decision.

English: The road was icy. **That's why** I decided to take the train.

Dutch: *Het was glad op de weg.* ***Daarom*** *besloot ik de trein te nemen.*

Analysis: The ice provided a *reason* for the speaker to make a conscious choice.

- Using the wrong marker would sound incorrect to a native speaker.

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

1. **Non-Volitional Cause (Event → Event):** a physical event is the direct cause of another event.

   English: The street was flooded **because of the fact that** the river burst its banks.

   Dutch: De straat was overstroomd **doordat** de rivier buiten haar oevers was getreden.

Analysis: The ice directly *caused* the sliding. The car did not make a choice.

2. **Volitional Cause (Event → Choice):** a person's deliberate action/reason is the cause of event.

   English: The street was blocked **because** the residents were holding a protest.

   Dutch: De straat was geblokkeerd **omdat** de bewoners protesteerden.

Analysis: The ice provided a *reason* for the speaker to make a conscious choice.

Rhetorical Structure Theory (RST)

(Mann and Thompson 1987), Web page

- Why bother with **volitionality**?
- **For tasks like translation or discourse relation parsing, ignoring volitionality leads to unnatural and incorrect language.**

- Also, RST schemas for languages with volitional markers reflect the same.
- For example, English RST has only `CAUSE`. But NId-RST has `NONVOLITIONAL-CAUSE` and `VOL-CAUSE`

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

- A large corpus where **relationships** between sentences/clauses have been annotated. It is based on the Wall Street Journal corpus.
- Unlike other frameworks like RST, PDTB doesn't force a tree structure on the entire text. Instead, it focuses on identifying **discourse connectives** and the parts of the text they link.

- The PDTB framework is **lexically grounded**, it starts by looking for specific words or phrases that signal a relationship.
- 3 key components for each relation:
  - **Discourse Connectives (or Alt Lex):** The word/phrase signaling the relation (e.g., *because, however, as a result, six year after*).
  - **Arg1:** The first part of the text in the relation.
  - **Arg2:** The second part of the text in the relation.

How do we organize longer pieces of text?

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

- PDTB organized the discourse relations into a **hierarchy (3 levels)**.

- PDTB sometimes calls its relations – senses.
- There are four top-level *classes*, 16 *types*, and 23 *subtypes* (not all types have subtypes).

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

Level-1	Level-2	Level-3
TEMPORAL	SYNCHRONOUS	–
	ASYNCHRONOUS	PRECEDENCE
		SUCCESSION
CONTINGENCY	CAUSE	REASON
		RESULT
		NEGRESULT
	CAUSE+BELIEF	REASON+BELIEF
		RESULT+BELIEF
	CAUSE+SPEECHACT	REASON+SPEECHACT
		RESULT+SPEECHACT
	CONDITION	ARG1-AS-COND
		ARG2-AS-COND
	CONDITION+SPEECHACT	–
	NEGATIVE-CONDITION	ARG1-AS-NEGCOND
		ARG2-AS-NEGCOND
	NEGATIVE-CONDITION+SPEECHACT	–
	PURPOSE	ARG1-AS-GOAL
		ARG2-AS-GOAL
COMPARISON	CONCESSION	ARG1-AS-DENIER
		ARG2-AS-DENIER
	CONCESSION+SPEECHACT	ARG2-AS-DENIER+SPEECHACT
	CONTRAST	–
	SIMILARITY	–
EXPANSION	CONJUNCTION	–
	DISJUNCTION	–
	EQUIVALENCE	–
	EXCEPTION	ARG1-AS-EXCPT
		ARG2-AS-EXCPT
	INSTANTIATION	ARG1-AS-INSTANCE
		ARG2-AS-INSTANCE
	LEVEL-OF-DETAIL	ARG1-AS-DETAIL
		ARG2-AS-DETAIL
	MANNER	ARG1-AS-MANNER
		ARG2-AS-MANNER
	SUBSTITUTION	ARG1-AS-SUBST
		ARG2-AS-SUBST

Table 1: PDTB-3 Sense Hierarchy. The leftmost column contains the Level-1 senses and the middle column, the Level-2 senses. For asymmetric relations, Level-3 senses are located in the rightmost column.

Note:
1. Senses like *joint* from PDTB 2 were removed (it was being used as a dumping ground for misc. relations), *list* was too similar to conjunction.
2. Rare senses were removed from PDTB 2.
3. Level-3 senses focus better on directionality.

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

- PDTB's **directionality** is in the sense of relation definition. It is better understood when looking at definitions such as PRECEDENCE vs. SUCCESSION

1. **Temporal.Asynchronous.Precedence** is used when the event described by **Arg1** precedes that described by **Arg2** (i.e., Arg1 << Arg2)

   [*A buffet was held in the museum*] (Arg1), then **[then they announced a 10-lap race]** (Arg2).

2. **Temporal.Asynchronous.Succession** is used when the event described by **Arg2** precedes that described by **Arg1** (i.e., Arg2 << Arg1)

   [Gates became a billionaire] (Arg1), six years after **[IBM adapted one of these versions in 1981.]** (Arg2).

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

- The PDTB framework is **lexically grounded**, meaning it starts by looking for specific words or phrases that signal a relationship.

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

- **Discourse Marker:** A discourse marker is a linguistic expression that signals the relationship between EDUs.
- **Discourse Relation:** The actual logical connection between two EDUs, such as **Cause**, **Contrast**, or **Elaboration**. This applies to RST, PDTB, SDRT, ISO….

"*I wanted to go to the park*, **but** it started to rain."

[CONTRAST]

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

Implicit Discourse Markers

- Not all coherence relations are marked by an **explicit discourse connective**. These are called **implicit discourse markers**.
- Oftentimes, PDTB also annotates pairs of neighboring sentences with **no explicit signal**.
- However, *the annotators were asked* to mark these implicit markers within sentences.

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

Alt Lex

- Discourse relations need not be marked by a single word. Sometimes, entire **phrases** act as linking units.
- These are alternative ways of lexically denoting these the relations (as opposed to simple words which can be limiting). These phrases are called **alternative lexicalizations (Alt Lex)**.

Penn Discourse TreeBank (PDTB 2.0, 3.0)

(Prasad et al 2007), (Prasad et al. 2018)

Contingency.Cause This tag is used when the situations described in Arg1 and Arg2 are causally influenced but are not in a conditional relation.

The label CONTINGENCY.CAUSE.REASON is used when Arg2 gives the reason, explanation or justification, while Arg1 gives its effect.

(68) Runways at San Francisco weren't damaged, *but traffic was being limited yesterday to 27 arrivals and 27 departures an hour – down from 33 to 45 an hour normally* – mainly because **the noise level in the control tower was overwhelming without the windows**, an FAA spokeswoman said. [wsj_1803]

19

(69) *But service on the line is expected to resume by noon today.* (Implicit=since) "**We had no serious damage on the railroad**," said a Southern Pacific spokesman. [wsj_1803]

(70) *By 11:59 p.m. tonight, President Bush must order $16 billion of automatic, across-the-board cuts in government spending to comply with the Gramm-Rudman budget law.* **The cuts are necessary because Congress and the administration have failed to reach agreement on a deficit-cutting bill**. [wsj_2384]

The label CONTINGENCY.CAUSE.RESULT is used when *Arg1* gives the reason, explanation or justification, while **Arg2** gives its effect.

(71) The bill would then declare *that the debt is equity* **and** therefore **isn't deductible**. [wsj_1822]

(72) *Now, though, enormous costs for earthquake relief will pile on top of outstanding costs for hurricane relief.* "**That obviously means that we won't have enough for all of the emergencies that are now facing us**, . . . ," Mr. Fitzwater said. [wsj_1824]

(73) "*We are going to explode lower*," says the flamboyant market seer, . . . (Implicit=so) **Anyone telling you to buy stocks in this market is technically irresponsible**. [wsj_0359]

The Explicitation Hypothesis (Blum-Kulka, 1986)

- **Explicitation** is a universal feature of translation. Translators (and MT models) tend to *spell things out* **more than the original author**, often to compensate for the loss of context that occurs when moving between linguistic and cultural systems or to reduce cognitive load (of understanding and interpreting at the same time).

- It is a type of Translation shift.
- Use case: Live translation at UN assembly.

Source:	Você também gosta dela?
Translation:	*So* you like her too?
Literal translation:	You like her too?

*Pym, A. (2005)

Here the translator has added the entirely optional adverbial *So*, knitting the discourse together in a way that the source does not

Explicitation and Implicitation of DMs

मीटिंग में मेरे **चाचाजी** भी थे।

[My **paternal uncle** was also in the meeting.]

"My **uncle** was also in the meeting."

- The English translation uses the general term **"uncle."** Specific information about which side of the family (paternal) is made implicit. In English-speaking contexts, this detail is unnecessary.

उसे चाय पसंद है, **लेकिन** मुझे कॉफ़ी पसंद है।

[He likes tea, **however**, I prefer coffee.]

He likes tea; I prefer coffee.

- Discourse markers get **implicitated** as well.

तुम मेहनत करो। तुम ज़रूर सफल होगे।

[You work hard. You will surely be successful.]

If you work hard, **then** you will surely be successful.

- And **explicitated**.

Explicitation and Implicitation of DMs

समुच्चय बोधक (Samyuchay bodhak):

Correlative conjunctions in English

We could **either** hike up the mountain **or** swim in the lake this afternoon.

Whether you bike **or** drive to work, you'll need to show your parking pass.

Not only did my boyfriend buy me a Nintendo Switch, **but** he also bought me a bunch of games!

Explicitation and Implicitation of DMs

हालाँकि वह थका हुआ था, **फिर भी** उसने काम पूरा किया।

[**Although** he was tired, **still** he completed the work.]

Although he was tired, he completed the work

- The English translation omits the second marker ("still" or "even so"). The word **"Although"** is strong enough on its own to signal the entire **Concession** relationship. The function of phir bhi becomes implicit, as its meaning is already contained within the concessive structure introduced by "Although."

समुच्चय बोधक (Samyuchay bodhak):

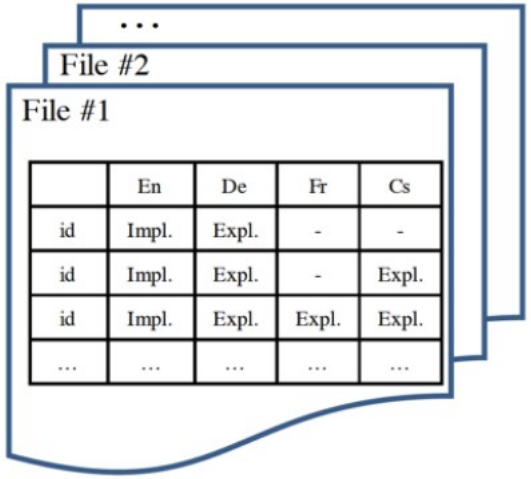
- **yadi... to (यदि... तो)**: If... then
  - *Yadi tum āoge, to main bhī āūṅgā.* (**If** you come, **then** I will also come.)
- **jab... tab (जब... तब)**: When... then
  - *Jab bārish hotī hai, tab mor nāchte haiṅ.* (**When** it rains, **then** the peacocks dance.)
- **jaisā... vaisā (जैसा... वैसा)**: As... so
  - *Jaisā karoge, vaisā bharoge.* (**As** you sow, **so** shall you reap.)

Implicit Discourse Relation Classification

- **Discourse Relation Classification** is the task of identifying the logical relationship (e.g., **Cause**, **Contrast**, **Elaboration**) between two units of text. For **implicit** relations, where no discourse marker is present, a system must automatically predict the correct relation by analyzing the content of the text spans themselves.

- The key in implicit discourse relation classification lies in extracting relevant information for the relation label from (the combination of) the discourse relational arguments. Informative signals can consist of surface cues, as well as the semantics of the relational arguments.

- **Backtranslation** is a technique where a text that has been translated from a source language (e.g., English) to a target language (e.g., Hindi) is then translated *back* to the original source language (English).
- Backtranslation can reveal implicit discourse relations.



	En	De	Fr	Cs
id	Impl.	Expl.	-	-
id	Impl.	Expl.	-	Expl.
id	Impl.	Expl.	Expl.	Expl.
...	...	...	...	...

Segmented Discourse Representation Theory (SDRT)

(Lascarides and Asher 2003)

- **SDRT - A Dynamic Model of Meaning**
- **What is it?** SDRT is a formal theory that models how we dynamically build a logical graph of a discourse, updating our understanding with each new sentence.
- **How it Works: The "Glue Logic"**
  - SDRT uses a logical inference system to determine the most plausible relationship between a new piece of information and the existing discourse context.
  - It considers semantics and world knowledge to attach new clauses with relations like **Explanation**, **Narration**, or **Contrast**.
- **Example: Building the Graph**
  - **Discourse:** "Max fell. John pushed him."
  - **SDRT's Process:**
    - "Max fell" is added as the first node.
    - "John pushed him" is introduced.
    - The "glue logic" infers that the most likely connection is **Explanation** and attaches the second clause to the first with that label.

Segmented Discourse Representation Theory (SDRT)

(Lascarides and Asher 2003)

The Right Frontier Constraint:

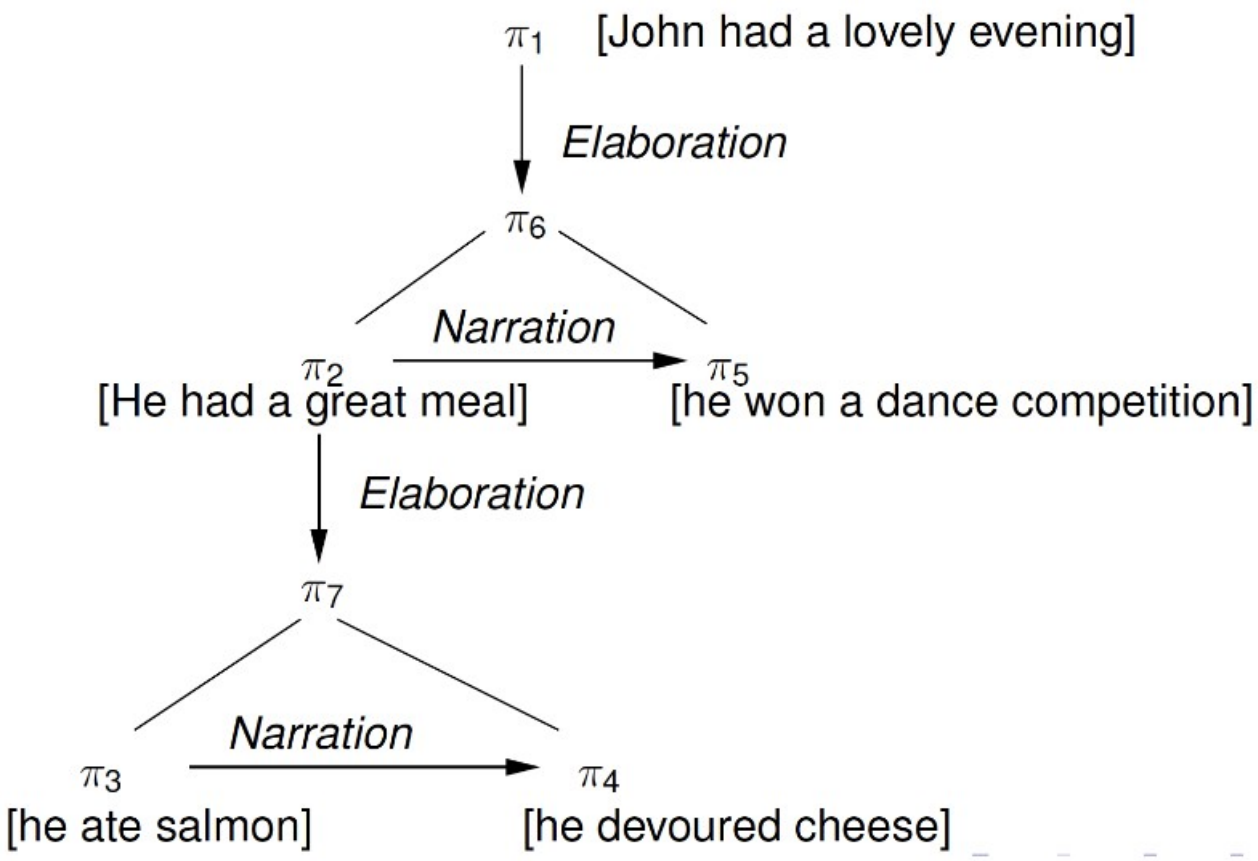
- Core rule governing how the discourse graph grows.
- New information can only attach to the currently "live" nodes on the graph's right edge. This models the natural flow of conversation and prevents nonsensical connections to long-past topics.

Why is SDRT Different?

- **Graph, Not a Tree:** Unlike RST's strict tree structure, SDRT's graph-based model is more flexible and can capture complex, non-hierarchical relationships.
- **Dynamic, Not Static:** It models the *process* of interpreting discourse in real-time, rather than just describing the final structure of a finished text.
- **Cognitively Focused:** It aims to be a plausible formal model of how the human mind actually processes and understands language as it unfolds.

Segmented Discourse Representation Theory (SDRT)

(Lascarides and Asher 2003)



Discourse relation classification

- **Discourse relation classification** is the task of automatically identifying and labeling the logical relationship (e.g., **Cause, Contrast, Elaboration**) between two spans of text. The goal is to predict the correct relation, especially when it is not explicitly marked by a discourse connective.

Discourse relation classification

The company's profits were high, but its stock price fell.

RST Analysis:

- [The company's profits were high] **(Nucleus)**
- [its stock price fell] **(Nucleus)**
- **Relation:** Contrast

PDTB Analysis:

- **Arg1:** The company's profits were high
- **Arg2:** its stock price fell
- **Connective:** <u>but, (explicit)</u>
- **Sense:** Comparison.Contrast

What are Classifiers?

F(x) = Y_i

x = EDU1, EDU2
Y_i ⊂ {Set of DRs}

Discourse relation classification

- **Rule-based Systems:** Rely on hand-crafted rules, primarily using discourse markers to classify relations. For example, a rule might state: "If the connective is *because*, label the relation as **Cause**.
  - While precise, these systems struggle with implicit relations where no marker is present. Also, it fails to work because the same word can be a discourse marker or a preposition or any other non discourse marker function.
  - First, mix the flour, **then** add the eggs. (*then* is a DM signaling **Temporal.Sequence** relation).
  - I wasn't ready **then**. (*then* is an adverb of time, not connecting two clauses).

- **Statistical / Machine Learning Systems:** Systems are trained on large annotated corpora (like the PDTB) to learn the patterns associated with different relations. They use features like word pairs, sentence structure, and deep learning embeddings (e.g., from models like XLM-RoBERTa) to predict the relation, even when it is implicit.

Georgetown Uni Multilayer corpus (GUM)

(Amir Zeldes 2017)

- Multilayered linguistic corpus.
- Also includes Discourse relations and markers.
- Multi domain
- News and Voyages are most similar in their distributions (less freqclass_Voy cooccur with less freq classes_News) possibly because writing style is similar, more `temporal` and evidence/.
- `Solutionhood` is used in argumentative texts where the interviewee is justifying his actions. Antithesis is used for contra-argumenting.

Text type	Source	Docs	Tokens
Academic writing	Various	18	17,169
Biographies	Wikipedia	20	18,213
CC Vlogs	YouTube	15	16,864
Conversations	UCSB Corpus	15	17,932
Courtroom transcripts	Various	9	11,148
Essays	Various	9	10,842
Fiction	Various	19	17,511
Forum	reddit	18	16,364

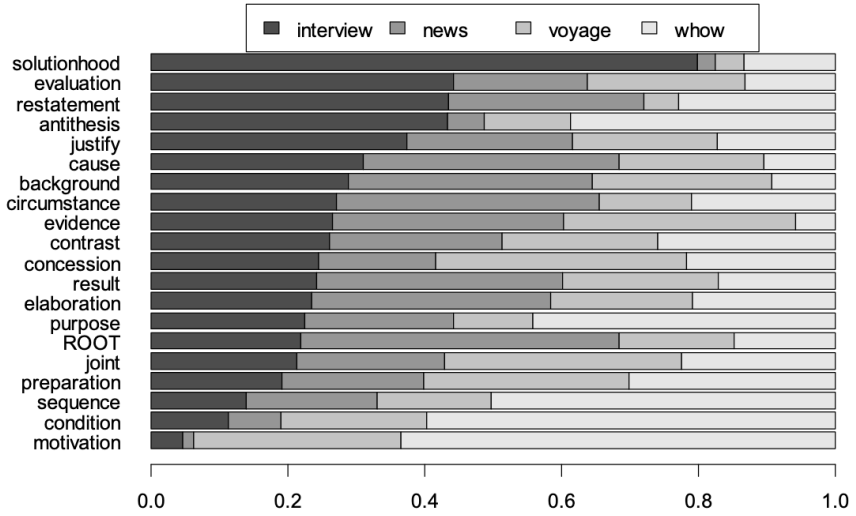


Figure 2. Relation proportions in four genres.

Hindi Dependency Treebank

(LTRC, IIITH)

- The treebank contains multiple layers of linguistic information
- **Dependency Structure:** Shows how words are grammatically related to each other.
- **Phrase Structure:** Shows how words combine to form phrases (like Noun Phrase, Verb Phrase).

- It also annotates Discourse Markers.

चूंकि यह एक बड़ा नीतिगत मामला है, इसलिये (ccof) इस पर मंत्रिमंडल की मुहर आवश्यक है।

- **Dependency Annotation:** HUTB's dependency structure, a marker like **isliye** (therefore) is tagged as a conjunction (ccof). Main clause ("the cabinet's seal is necessary") to the subordinate clause ("Since this is a big policy matter") that underlies the **Cause-Result** discourse relation.
- While HUTB doesn't explicitly label the sentence with "Cause-Result," the grammatical annotation of the markers facilitates identification.

CODI

(Shared Task)

CODI (Cross-lingual Discourse-relation Identification) task is a shared task on developing systems to identify discourse relations in multiple languages.

There are 3 tasks in this discourse analysis shared task:

- **Discourse Segmentation:** Identifying the elementary discourse units (EDUs).
- **Connective Identification:** Finding the words that explicitly signal a discourse relation.
- **Relation Classification:** Predicting the logical relationship between discourse units.

The company's profits were high, (so, implicit) it gave bonuses.

- A sequence labeling task of **DS**. Output:
  - *Arg1*: The company's profits were high
  - *Arg2*: so it gave bonuses
- A sequence labelling of **CI** if connective is explicit. Output:
  - *Connective*: so
- A classification task of **RC**. Output:
  - *Relation*: Cause.Result

CODI

(Shared Task)

CODI (Cross-lingual Discourse-relation Identification) task is a shared task on developing systems to identify discourse relations in multiple languages.

1. Schema Heterogeneity: Different discourse corpora are often annotated using different schemas (e.g., PDTB, RST). This creates significant challenges:

- The schemas may have entirely different sets of relation labels.
- Even when two schemas use the same relation name (e.g., "Contrast"), what qualifies for that label can have subtle but important differences.

2. Domain and Genre Variation: The datasets in CODI span multiple domains (e.g., news, legal texts, online reviews). This impacts the task in two ways:

- **Relation Distribution:** The frequency of different relations changes dramatically across domains.
- **Model Bias:** A model trained heavily on one domain may perform poorly on another because it has learned a biased distribution of relations.
- It is a **domain drift (data drift)**. Can occur in essay grading, making models even time series analysis.

3. Cross-Lingual Transfer: Building a single model for multiple algnauges is incredibly difficult:

- **Propensity for Marking:** Languages mark relations with differing frequency.
- **Model Requirements:** This challenge necessitates either massive, universal LLMs.

4. Varying dataset sizes: Only few of the datasets have sizes larger than 10K+ relations. Models overfit on smaller datasets.

5. Class-imbalance: Models overfit on most frequent classes.

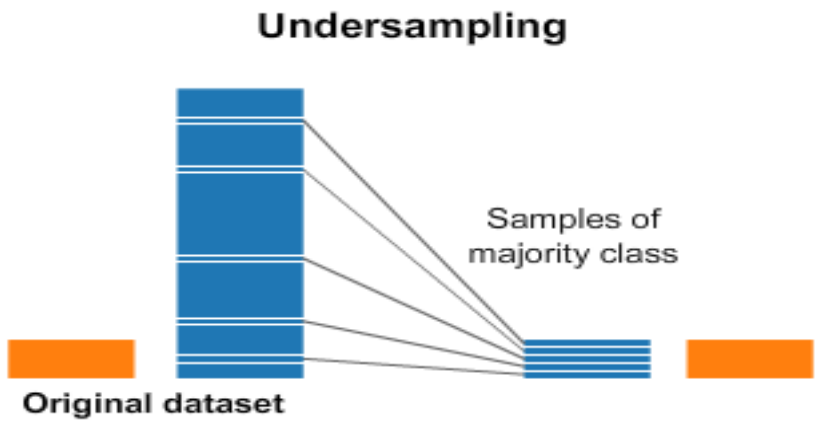
CODI

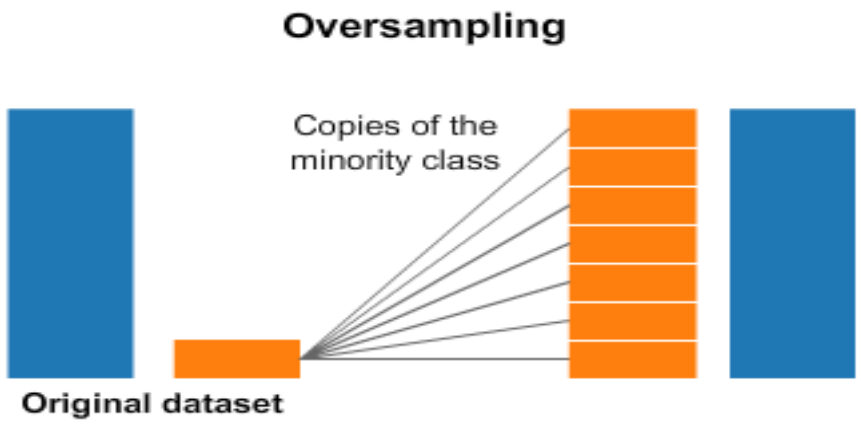
(Shared Task)

- **Undersampling:** Balances the dataset by randomly **removing** examples from the **larger, majority group** until it's the same size as the smaller group.

- **Oversampling:** Makes exact copies of the examples in **the smaller**, minority group to increase its size. Comparable to the larger set. How much over or under sample you want to do (the degree) can be treated as a hyperparameter and tuned on dev set.

- **SMOTE (Synthetic Minority Over-sampling Technique):** This is a smarter oversampling technique that creates new, *synthetic* examples of the minority group that are similar to, but not exact copies of, the originals.





CODI

(Shared Task)

corpus	lang	framework	rel_types	rels
ces.rst.crdt	ces	rst	17	1,249
deu.pdtb.pcc	deu	pdtb	11	2,109
deu.rst.pcc	deu	rst	16	2,882
eng.dep.covdtb	eng	dep	11	4,985
eng.dep.scidtb	eng	dep	14	9,903
eng.erst.gentle	eng	erst	17	2,552
eng.erst.gum	eng	erst	17	30,747
eng.pdtb.gentle	eng	pdtb	12	786
eng.pdtb.gum	eng	pdtb	13	13,879
eng.pdtb.pdtb	eng	pdtb	13	47,792
eng.pdtb.tedm	eng	pdtb	13	529
eng.rst.oll	eng	rst	17	2,751
eng.rst.rstdt	eng	rst	17	19,778
eng.rst.sts	eng	rst	17	3,058
eng.rst.umuc	eng	rst	15	4,997
eng.sdrt.msdc	eng	sdrt	10	27,848
eng.sdrt.stac	eng	sdrt	11	12,271
eus.rst.ert	eus	rst	16	3,632
fas.rst.prstc	fas	rst	14	5,191
fra.sdrt.annodis	fra	sdrt	12	3,321
fra.sdrt.summre	fra	sdrt	0	0
ita.pdtb.luna	ita	pdtb	11	1,525
nld.rst.nldt	nld	rst	16	2,264
pcm.pdtb.disconaija	pcm	pdtb	13	9,903
pol.iso.pdc	pol	iso	12	8,543
por.pdtb.crpc	por	pdtb	12	11,327

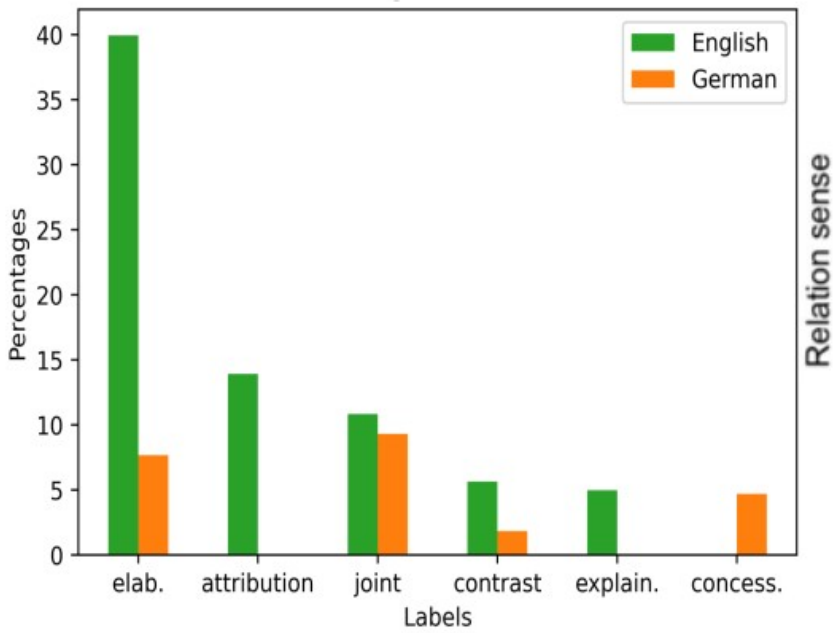


Fig 1. Label distribution across English and German RST datasets (DISRPT task)

Discourse Markers as Predictors of Code Switching

- **Code-switching** is when a speaker alternates between two or more languages or language varieties **between sentences or clauses**. It's a shift that occurs at a grammatical boundary.

I have to finish my work. *Phir main free ho jaunga.* (Then I will be free.)

- Complete a full sentence/phrase/chunk in English before switching to a full sentence in Hindi.

- **Code-mixing** is when a speaker mixes words or phrases from two or more languages **within a single sentence or clause**. This involves embedding elements of one language into the grammatical structure of another.

My mom is very *gussa* right now. (My mom is very angry right now.)

- Here, the Hindi word "gussa" is inserted directly into the middle of an English sentence.

Discourse Markers as Predictors of Code Switching

- **Intersentential switching** occurs *outside* the sentence /clause clause level (i.e. at sentence or clause boundaries). In Assyrian-Eng switching :

  Ani wideili. What happened? (*Those, I did them.* What happened?).

- **Intra-sentential switching** occurs *within* a sentence/clause. In Spanish-En switching:

  La onda is to fight *y jambar.* (*The latest fad* is to fight *and steal.*)

- **Matrix Language** (ML): main or dominant language
- **Embedded Language** (EL): **secondary language** from which words/phrases are inserted into the structure of the ML.

  My mom is very $_{[ML]}$ *gussa* $_{[EL]}$ right now. $_{[ML]}$

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  నేను నీకు [ML] *call* [EL] చేస్తాను.[ML] (I, to you *call* will do -> I will call you.)

  En: object ("you") is part of the VP ("will call you"), in Telugu: object (nīku) stands alone as its own NP before the verb.

Discourse Markers as Predictors of Code Switching

- Spanish-English bilinguals highlights this pattern. A study by Gumperz (1982) provides a classic example:

  I'm not going to the party, **y'know**, *no tengo ganas*. (I don't like it)

- The ***Closed-class Constraint***, posits that closed class items (pronouns, prepositions, conjunctions, etc.) cannot be switched.
- Research has largely moved beyond the strict Closed-class Constraint. While Open-class items (nouns, verbs) are switched more frequently, even closed-class (prepositions, conjunctions) items are switched, including discourse markers.
- Czech-(austro)English bilinguals: J. Hlavac (2003) showed that fillers (yeah, uh-huh) and DMs are code mixed.

  tamo mi nije bilo bas ˇnajbolje (I didn't have the best of times there).. **so**.. *Ovaj* (this, um) .. *morala sam onaj napravit malo prijatelji*(I had to make a few friends)

- **DMs** often act as **pivots or triggers** for a language switch. Their position at the beginning of a clause, a bridge onto the next phrase, makes them a natural point to shift from one language to another.
- **EDU boundaries** are another natural point for switching.

[Clause in Language A] + [DM from Language A] + **[Clause in Language B]**

[Clause in Language A] + **[DM from Language B]** + **[Clause in Language B]**

[Clause in Language A] + **[DM from Language B]** + [Clause in Language A]*rare (3 Medial:21 Initial/Final)

Discourse Markers as Predictors of Code Switching

- 4:19 – Codeswitching after DM. Arg1-En, DM-En, Arg2-Hi
- 4:31 – Codeswitching after AltLex. Arg1-Hi, AL-Hi, Arg2-En
- 4:31 – Codeswitching at EDU boundary.
- 8:36 - Codeswitching after DM. Arg1-En, DM-En, Arg2-Hi
- https://www.youtube.com/watch?v=ba9TJFBdFQ8&t=3129s

- 8:35 – Codemixing only at AltLex. Arg1-Hi, AL-En, Arg2-Hi
- 9:40 – Codemixing only at DM(but). Arg1-Hi, DM-En, Arg2-Hi
- 11:03 – Codemixing only at DM(ki). Arg1-En, DM-Hi, Arg2-En
- *Codemixing because the embedding matrix belongs to Arg1 and Arg2.

- https://www.youtube.com/watch?v=x0cB71miXmw

Individual Differences in DR Classification

(Scholman et al 2022)

- **Individual Differences:** are the psychological traits and characteristics that distinguish one person from another.

- **Learning from Disagreement: Classifiers are trained on majority labels.**
- Human uncertainty makes classification more robust. (Peterson, Battleday 2019).
- When you are uncertain, how do you make a decision?
- *A four legged blob is likely to be a horse/dog and not an airoplane.*

Individual Differences in DR Classification

(Scholman et al 2022)

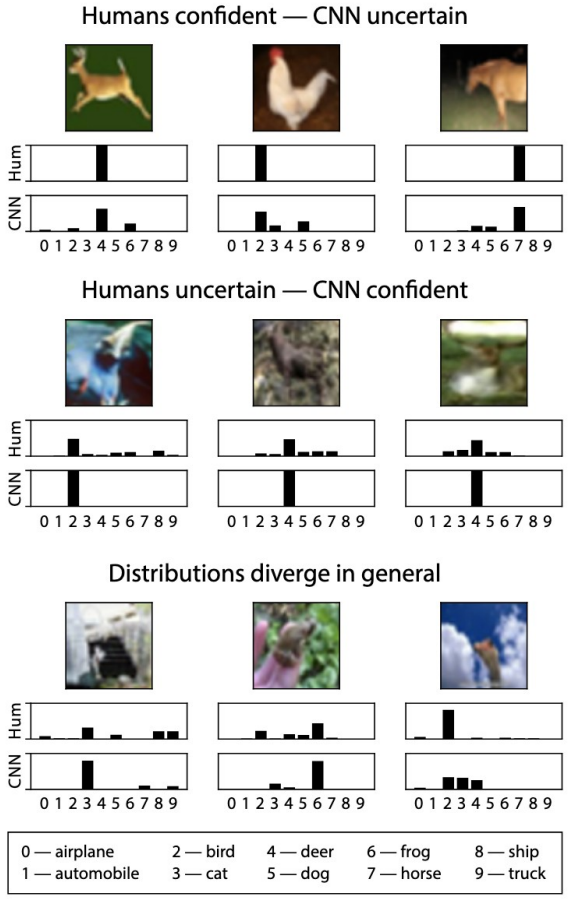
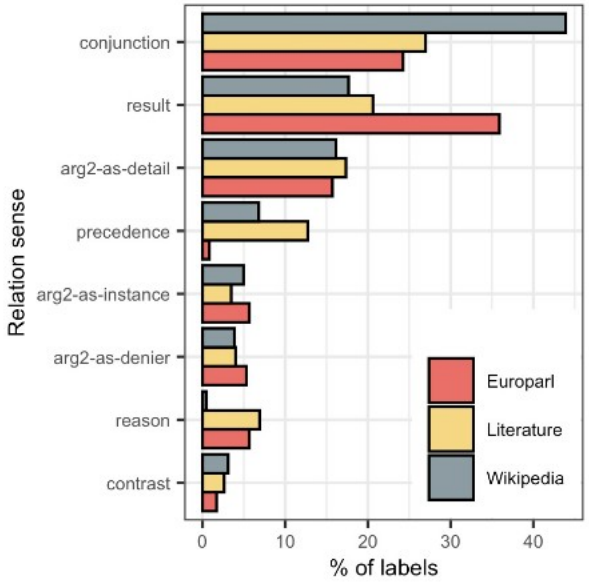


Figure 1: CIFAR10 images for which humans and our best traditionally-trained CNN (Shake-Shake [11]) agree in their top guess, but systematically differ over other choices.

Individual Differences in DR Classification

(Scholman et al 2022)

Clear differences in relational distribution between genres:



- CONJUNCTION most prevalent in Wikipedia
- RESULT relations occur more in Europarl
- Most PRECEDENCE relations in literature

Highlights the importance of taking genre effects into consideration

Individual Differences in DR Classification

(Scholman et al 2022)

- **DiscoGeM:** A **crowdsourced, genre-mixed, and multilingual corpus** of **implicit discourse relations**. Its key feature is capturing the ambiguity of discourse by providing a *distribution* of possible relation labels from multiple annotators for each instance, rather than a single definitive label.

- **Individual Differences: Individual differences** are the psychological traits and characteristics that distinguish one person from another. A person will predict relations slightly different from others within reasonable constraints.

- **Discourse Relation Classification:** The task is to automatically identify the sense(s) of an **implicit discourse relation**. A key challenge highlighted by DiscoGeM is that systems should ideally predict a **probability distribution** over multiple possible labels rather than just one, to account for inherent ambiguity.

Georgetown Uni Multilayer corpus (GUM)

(Amir Zeldes 2017)

- Multilayered linguistic corpus.
- Also includes Discourse relations and markers.
- Multi domain
- News and Voyages are most similar in their distributions (less freqclass_Voy cooccur with less freq classes_News) possibly because writing style is similar, more `temporal` and evidence/.
- `Solutionhood` is used in argumentative texts where the interviewee is justifying his actions. Antithesis is used for contra-argumenting.

Text type	Source	Docs	Tokens
Academic writing	Various	18	17,169
Biographies	Wikipedia	20	18,213
CC Vlogs	YouTube	15	16,864
Conversations	UCSB Corpus	15	17,932
Courtroom transcripts	Various	9	11,148
Essays	Various	9	10,842
Fiction	Various	19	17,511
Forum	reddit	18	16,364

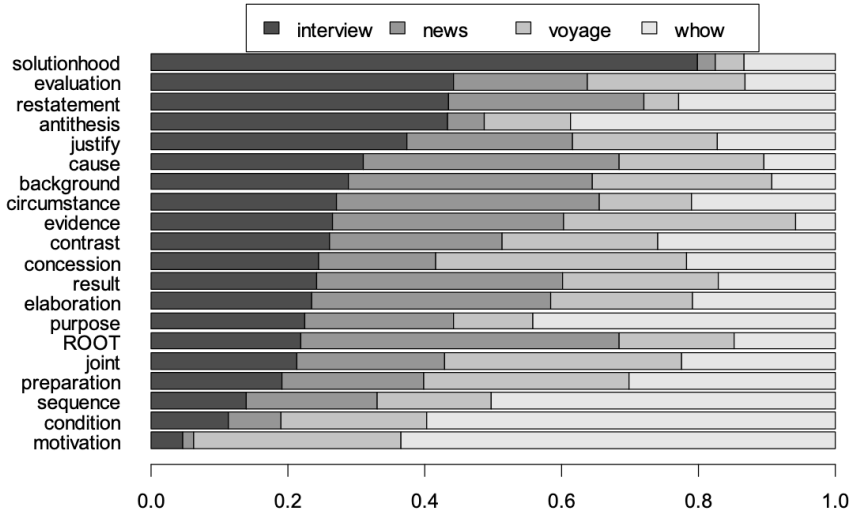


Figure 2. Relation proportions in four genres.

Domain impacts Discourse relations

- Domain have different **distributions for different grammatical categories** because of writing style differences.
- This impacts DR distributions across domains.

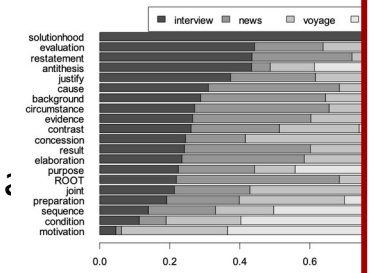


Figure 2. Relation proportions in four genres.

- Not just raw probability distributions but even co-occurences a domain differences.

- **NPMI (Normalized Pointwise Mutual Information)** is a score that measures how often two things appear together compared to how often they are expected to appear together by pure chance.

$$NPMI(x,y) = \frac{\log\left(\frac{P(x,y)}{P(x)\cdot P(y)}\right)}{-\log(P(x,y))}$$

Europarl	CNJ	RES	REA	DET	INS	CNC	CNT	PRE
CNJ		.184	.063	.053	.010	.008	-.196	-.209
RES	.214		-.061	.018	.066	.175	-.025	-.172
REA	.024	-.033		.180	.103	.072	.113	-.185
DET	.057	.035	.042		.262	-.111	-.114	-.381
INS	.001	-.072	.093	.230		-.004	.000	.000
CNC	.029	.097	.001	-.118	.025		.217	.000
CNT	-.320	.005	-.047	.000	-.021	.235		.000
PRE	-.245	-.104	.000	.000	.000	-.176	-.062	
ALN	-.097	.234	.008	-.016	.033	.025	-.087	-.800

Novel	CNJ	RES	REA	DET	INS	CNC	CNT	PRE
CNJ		-.045	.006	.198	-.087	-.093	.018	.192
RES	.011		-.097	-.042	-.129	-.064	-.080	.333
REA	-.042	-.080		.171	.056	-.134	.000	-.048
DET	.144	-.172	.198		.227	-.063	-.025	-.132
INS	-.063	-.162	-.054	.153		-.110	.000	-.076
CNC	-.086	.019	.000	-.131	.000		.124	.054
CNT	.060	-.063	.032	.083	-.032	.141		-.079
PRE	.109	.246	.000	-.164	-.025	.087	.000	
ALN	.008	-.128	.085	.007	-.141	-.031	.047	.175

Wikipedia	CNJ	RES	REA	DET	INS	CNC	CNT	PRE
CNJ		.097	.000	.257	.091	-.185	.197	.208
RES	.153		.000	.004	-.017	-.137	.000	.196
REA	.000	.000		-.148	.000	.000	.000	.000
DET	.209	-.245	.000		.371	.000	.000	.000
INS	.130	.000	.000	.306		.101	.157	.000
CNC	-.021	-.096	.000	.000	.000		.302	.122
CNT	.103	-.071	.000	.004	.000	.385		.000
PRE	.186	.108	.000	-.140	.000	.220	.120	
ALN	.337	-.291	-.740	-.055	-.011	-.084	.059	-.060

Figure 1: NPMI per genre of co-occurrences of the most frequent relation senses. The x-axis represents the TOP1 label; the y-axis the TOP2 label.

Domain impacts Discourse relations

- Relations in Wikipedia are more likely to co-occur with **CONJUNCTIONS**, and **CONJUNCTIONS** are also more likely to appear alone. The study directly attributes this to the **"expository nature"** of Wikipedia.

- **RESULT** and **PRECEDENCE** relations are much more likely to co-occur in novels than in the Europarl data. This reflects the **narrative writing style** of novels.

- In the Europarl data, **RESULT** relations are more likely to appear alone (as a single label) compared to other genres. The study connects this to the **"argumentative nature"** of the political genre. In an argument or debate, a speaker often makes a point and then states its direct, standalone consequence to persuade the audience.

- *Models that learn multi-label distributions are better than one-hot label classifiers.* Imagine penalizing a model for predicting CONTRAST vs. REASON when the gold label is CAUSE.

Europarl	CNJ	RES	REA	DET	INS	CNC	CNT	PRE
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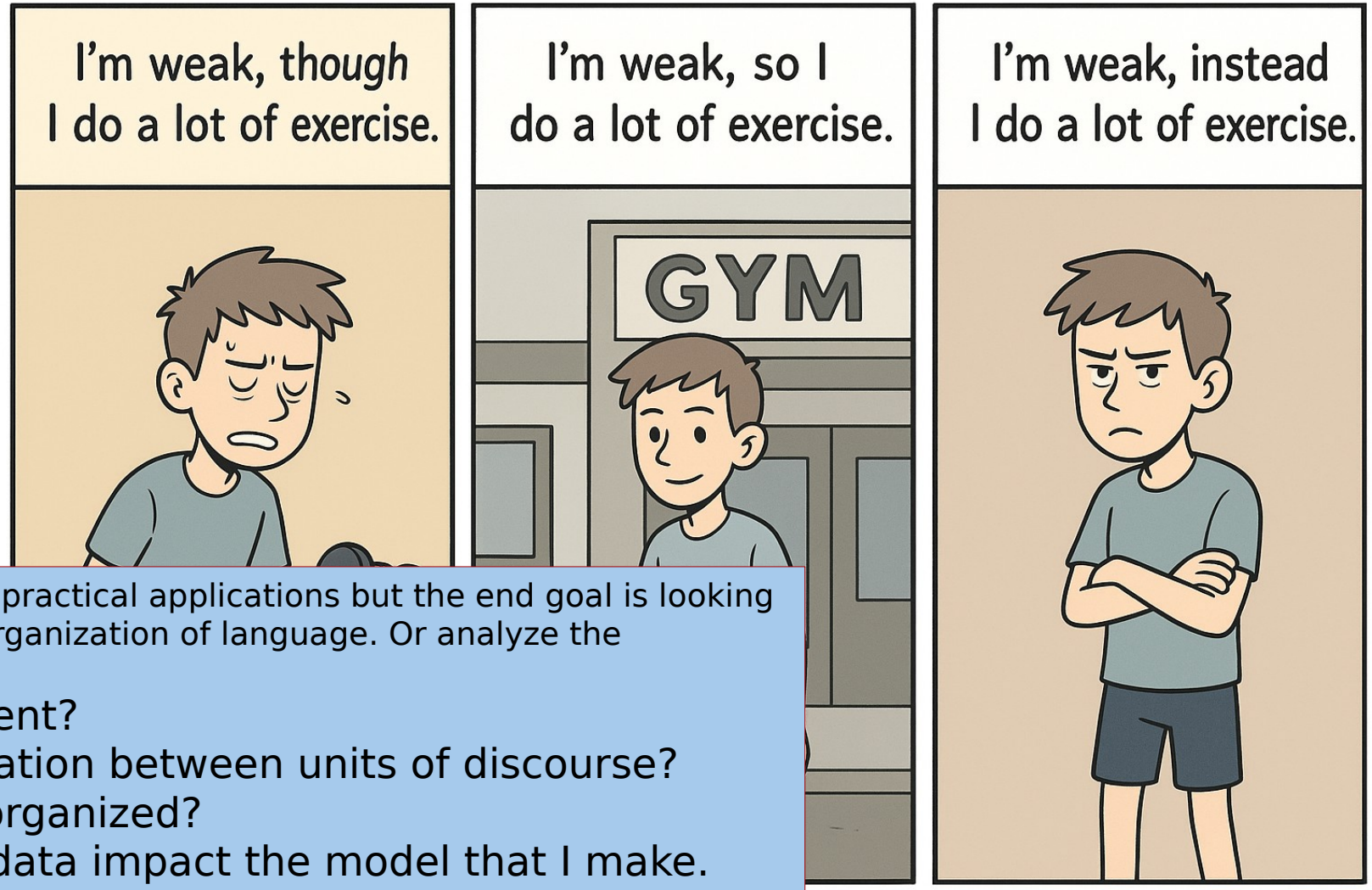
DiscoFlan
(Anuranjana 2023)

- Jump to slides
- What will I change – use a classification head :P

Motivation: DMs = Coherence ≃ *Intent



These were individual practical applications but the end goal is looking at the structure and organization of language. Or analyze the discourse.

- What is my intent?
- What is the relation between units of discourse?
- How are they organized?
- How does the data impact the model that I make.
- What *information* is carried in these units?

References

*Your class notes!

Computational concepts

- Jurafsky
- DiscoGem
- DiscoFlan
- GUM
- Transfer learning
- Class imbalance
- Uncertainty in classification

Major linguistic theories

- PDTB 2 manual
- PDTB 3 manual
- RST paper by Mann and Thompson
- SDRT